

Assignment No. 8

Title: Google App Engine.

Problem Statement: Installation and configure Google App Engine.

Objective: Installation and configure Google App Engine.

Outcome: Installation and configure Google App Engine.

Theory:

The Google App Engine (GAE) is a Platform as a Service (PaaS) offering from Google Cloud Platform (GCP). In a PaaS model, developers focus only on writing and deploying application code, while the cloud provider manages infrastructure such as virtual machines, operating systems, scaling, load balancing, and security updates.

App Engine provides two environments: Standard and Flexible.

In this experiment, the Standard environment was used, which runs applications in a secure sandbox and automatically scales based on traffic.

The developer does not need to configure servers manually.

Unlike Infrastructure as a Service (IaaS) platforms such as AWS EC2, where the user must install software, configure web servers, and manage ports, App Engine abstracts all infrastructure layers. The application is deployed using the `gcloud app deploy` command, and Google automatically provisions the required runtime environment.

Each App Engine project has:

- A project ID
- A region (selected during initialization)
- A default service URL (`project-id.region.r.appspot.com`)

The application configuration is defined using an `app.yaml` file. This file specifies the runtime environment (e.g., Python 3.11) and the entry point command used to run the application.

Dependencies are defined in a `requirements.txt` file.

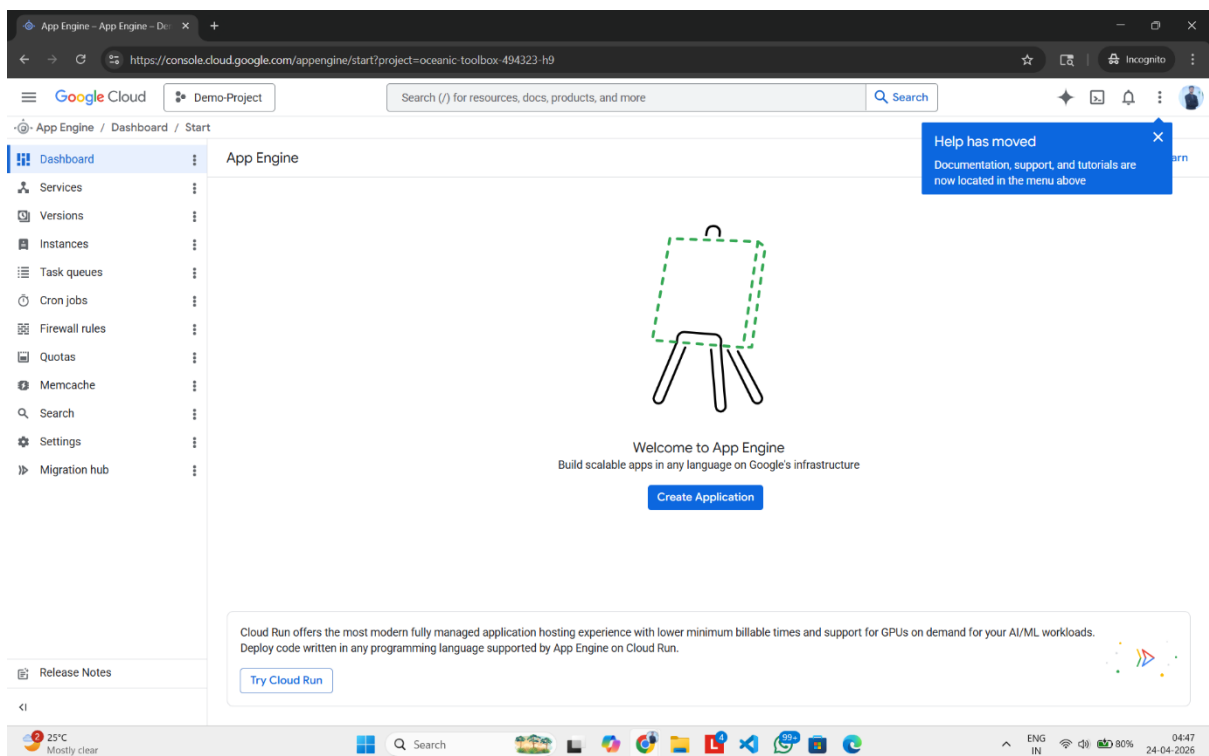
During deployment, Google Cloud:

1. Uploads the source code.

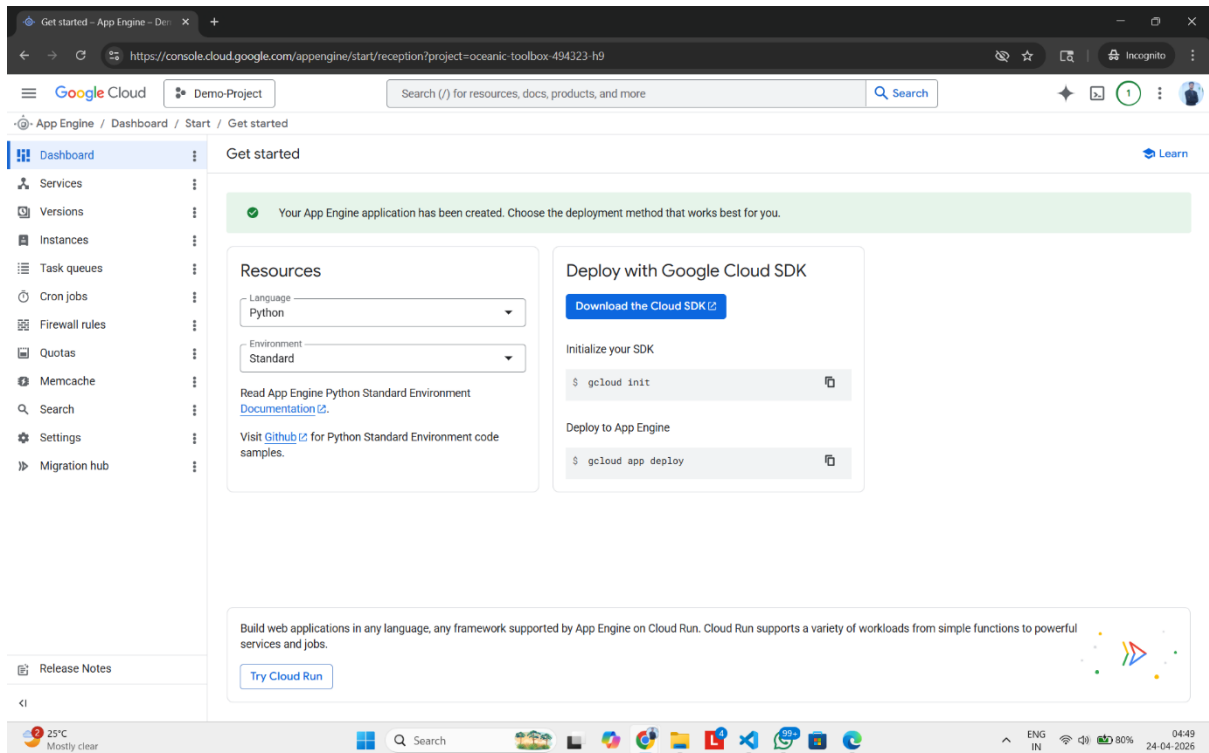
2. Installs dependencies.
3. Creates a containerized runtime environment.
4. Starts the application using the specified entry point.
5. Assigns a public URL.

Procedure:

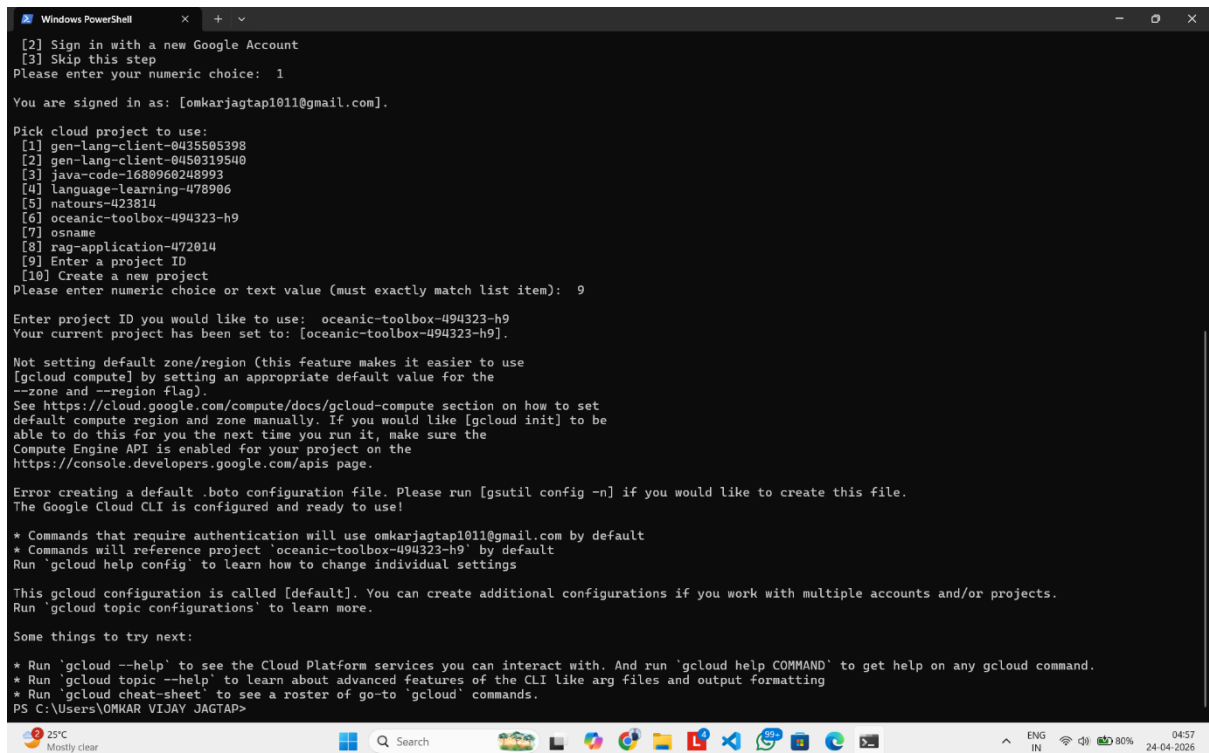
Step 1: Go to Google Console and Search App Engine



Step 2: Select the region and resources



Step 3: Come to terminal and enter 'gcloud init'



Step 4: Verify Config using 'gcloud config list'

```
Windows PowerShell
PS C:\Users\OMKAR VIJAY JAGTAP> gcloud config list
[accessibility]
screen_reader = False
[core]
account = omkarjagtap1011@gmail.com
disable_usage_reporting = true
project = oceanic-toolbox-494323-h9

Your active configuration is: [default]
PS C:\Users\OMKAR VIJAY JAGTAP>
```

Step 5: Deploy the app using 'gcloud app deploy'

```
Windows PowerShell
PS C:\Users\OMKAR VIJAY JAGTAP\my-app> gcloud app deploy
WARNING: You might be using automatic scaling for a standard environment deployment, without providing a value for automatic_scaling.max_instances. Starting from March, 2025, App Engine sets the automatic scaling maximum instances default for standard environment deployments to 20. This change doesn't impact existing apps. To override the default, specify the new max_instances value in your app.yaml file, and deploy a new version or redeploy over an existing version. For details on max_instances, see https://cloud.google.com/appengine/docs/standard/reference/app-yaml.md#scaling_elements.

Services to deploy:

descriptor:      [C:\Users\OMKAR VIJAY JAGTAP\my-app\app.yaml]
source:          [C:\Users\OMKAR VIJAY JAGTAP\my-app]
target project:  [oceanic-toolbox-494323-h9]
target service:  [default]
target version:  [20260424t050907]
target url:       [https://oceanic-toolbox-494323-h9.el.r.appspot.com]
target service account: [oceanic-toolbox-494323-h9@appspot.gserviceaccount.com]

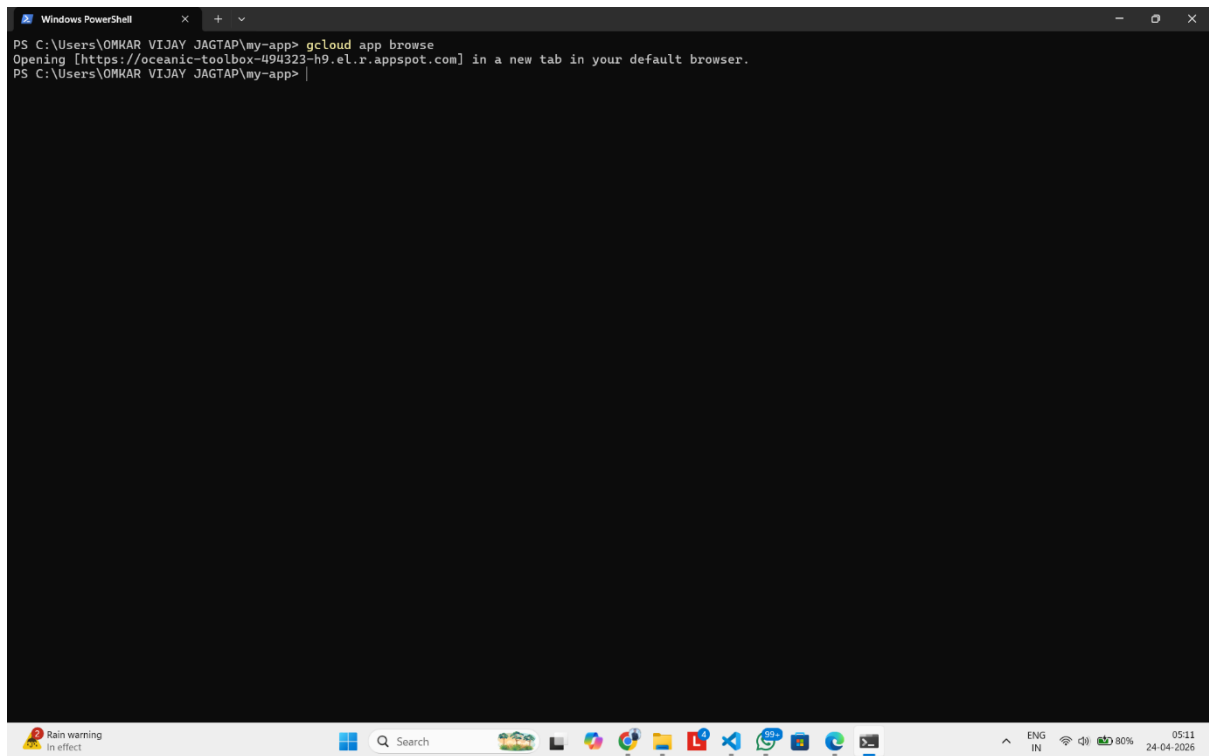
Do you want to continue (Y/n)? y

Beginning deployment of service [default]...
#=====
# Uploading 1 file to Google Cloud Storage      ==#
#=====
File upload done.
Waiting for operation [apps/oceanic-toolbox-494323-h9/operations/73f88d5e-e877-4f79-9750-f6677901cab3] to complete...done.
Setting traffic split for service [default]...done.
Deployed service [default] to [https://oceanic-toolbox-494323-h9.el.r.appspot.com]

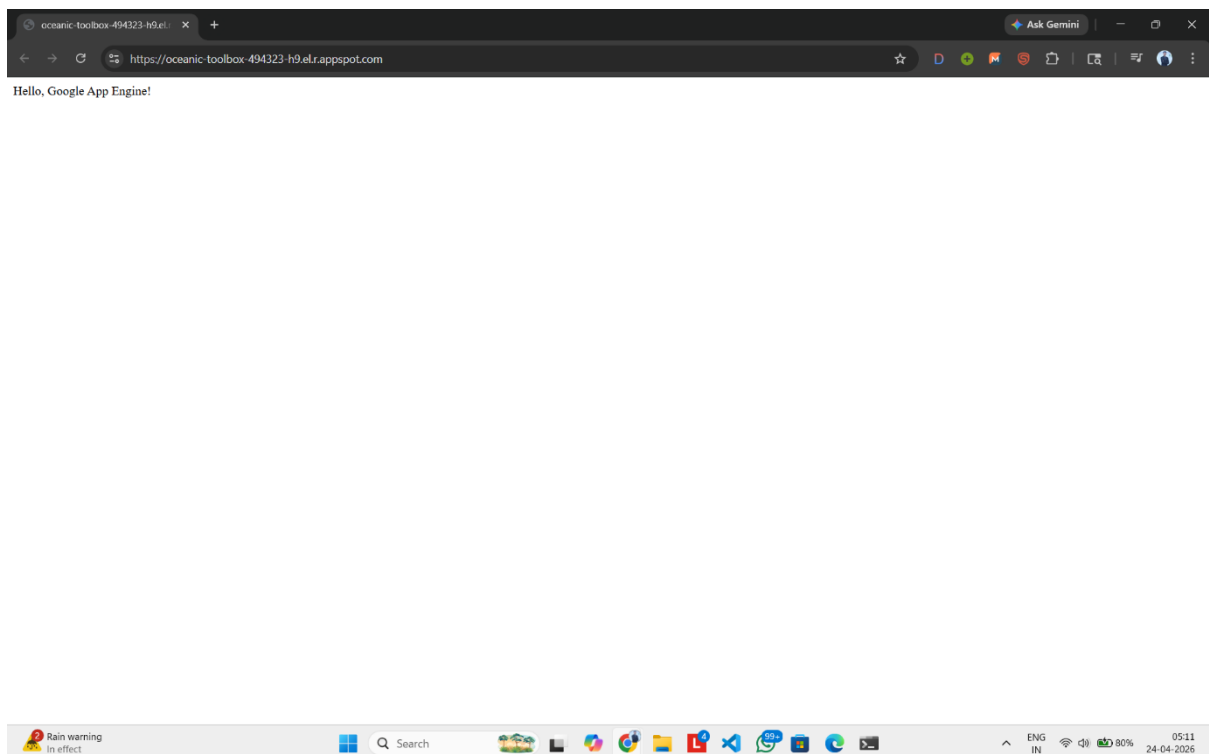
You can stream logs from the command line by running:
$ gcloud app logs tail -s default

To view your application in the web browser run:
$ gcloud app browse
PS C:\Users\OMKAR VIJAY JAGTAP\my-app> |
```

Step 6: Launch app using 'gcloud app browse'



```
Windows PowerShell
PS C:\Users\OMKAR VIJAY JAGTAP\my-app> gcloud app browse
Opening [https://oceanic-toolbox-494323-h9.el.r.appspot.com] in a new tab in your default browser.
PS C:\Users\OMKAR VIJAY JAGTAP\my-app> |
```



Conclusion: Google App Engine was successfully installed and configured. A simple web application was deployed using the Python runtime. The

experiment demonstrated the functioning of the Platform as a Service model and highlighted the difference between PaaS and IaaS cloud services